

VOCAL PATH: A CLOUD-ENABLED INTELLIGENT SYSTEM FOR ADAPTIVE SPEECH THERAPY IN STUTTERING

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Project Proposal Report

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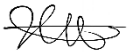
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Declaration

We declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Stuttering is a speech fluency disorder characterized by involuntary disruptions in the smooth, rhythmic flow of speech, including sound repetitions, prolongations, and speech blocks. Despite its prevalence, stuttering often remains underrecognized and underdiagnosed, primarily due to limited public awareness and the lack of accessible, standardized screening methodologies. This study proposes a systematic approach for the early detection and longitudinal monitoring of stuttering in children aged 5-10 years. Given that stuttering manifests across a spectrum of severities and can profoundly impact a child's communicative competence and psychosocial well-being, early intervention is critical. The proposed framework employs a structured and child-centered speech assessment protocol designed to evaluate speech fluency, temporal rhythm, and articulatory consistency in naturalistic speaking contexts. Speech data are collected and analyzed to identify quantifiable disfluency patterns indicative of stuttering. Upon detection, the system provides individualized feedback and recommends targeted fluency intervention exercises, enabling continuous progress tracking over time. The integration of children, parents, and speech-language pathologists within a unified monitoring platform facilitates collaborative therapy management, offering transparent progress visualization and data-driven outcome evaluation. This approach aims to enhance early identification, support evidence-based intervention, and ultimately improve communicative outcomes for children affected by stuttering..

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List Of Abbreviations

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **API** – Application Programming Interface
- **ASR** – Automatic Speech Recognition
- **SLP** – Speech-Language Pathologist
- **SLT** – Speech-Language Therapist
- **MFCC** – Mel-Frequency Cepstral Coefficients
- **KNN** – K-Nearest Neighbors
- **RMSE** – Root Mean Square Error
- **MAE** – Mean Absolute Error
- **IDE** – Integrated Development Environment
- **SDK** – Software Development Kit
- **CLI** – Command Line Interface
- **IEEE** – Institute of Electrical and Electronics Engineers
- **AIS** – Association for Information Systems
- **IET** – Institution of Engineering and Technology
- **CSSL** – Computer Society of Sri Lanka
- **ACS** – Australian Computer Society

1. Introduction

Stuttering is a speech fluency disorder characterized by involuntary disruptions in the smooth flow of speech, despite normal language comprehension and intact motor abilities required for articulation [1][2]. Although stuttering is not associated with reduced intelligence, children with this condition often struggle with maintaining fluent, rhythmical, and effortless speech. The challenges associated with stuttering may include:

- Frequent sound or syllable repetitions, such as “b-b-ball.”
- Prolongations of sounds, where speech is stretched unnaturally (e.g., “ssssun”).
- Blocks or pauses, in which speech is temporarily halted despite effort to speak.
- Irregular speech rhythm and rate, disrupting the natural prosody of spoken language.
- Visible physical tension or struggle behaviors, including eye blinking, facial movements, or jaw tremors during speech attempts.
- Avoidance of certain words or speaking situations, stemming from anticipation of stuttering moments.
- Increased speech disfluency under emotional stress or time pressure.
- Negative emotional responses, such as frustration or anxiety associated with speaking.

It is important to recognize that every child who stutters experiences the condition differently. Some may exhibit mild and occasional disfluencies, while others face frequent, severe disruptions that significantly impact daily communication [3][4]. Moreover, stuttering exists along a continuum, with varying degrees of severity that can influence academic engagement, social participation, and emotional well-being. A comprehensive understanding of stuttering is essential to fostering supportive educational and therapeutic environments and to implementing individualized intervention strategies. Early detection and evidence-based interventions can markedly enhance a child’s fluency, communication confidence, and overall quality of life [5][6].

Within these diverse manifestations, the present research focuses specifically on the early identification of speech disfluency patterns and rhythm disruptions in children aged 5–10 years, thereby enabling timely diagnosis and the formulation of targeted fluency enhancement therapies.

2. Background & Literature Survey

2.1. Background

Stuttering is a speech fluency disorder that affects a child's ability to produce smooth, continuous, and rhythmical speech. Children with this condition know exactly what they want to say, but their speech is involuntarily interrupted by repetitions, prolongations, or blocks of sounds and syllables. These disruptions occur even though the child has normal muscle strength and language comprehension abilities [1], [2]. Importantly, stuttering does not reflect reduced intelligence or cognitive ability; children who stutter are just as capable as their peers in understanding and expressing ideas [3].

Stuttering can range from mild to severe, but it often becomes most apparent in children between the ages of 5- 10 when fluent communication is increasingly important for academic performance and social development [4]. Children who stutter may repeat sounds or words ("b-b-ball"), stretch out certain sounds ("ssssun"), or experience brief, involuntary pauses where no sound comes out at all. These speech disruptions can vary across situations, often becoming more pronounced during stressful or high-pressure contexts. Additionally, irregular rhythm, misplaced stress, and speech tension may further interfere with the natural flow of speech [5].

A distinctive feature of stuttering is the presence of physical struggle or tension as the child attempts to speak. This may include visible signs such as eye blinking, lip tremors, or facial tension. These behaviors often reflect the child's effort to overcome blocks or initiate speech, and they can contribute to emotional distress or frustration [6]. Furthermore, longer or more complex words and sentences tend to trigger greater speech disfluency, making classroom communication and social interaction more challenging. For instance, a child may fluently say a simple word like "dog" but encounter difficulty with longer phrases such as "my favorite animal is a dog."

Early identification of stuttering symptoms is critical because timely intervention can significantly improve fluency, communication confidence, and overall quality of life. Recognizing stuttering as a disorder of speech fluency rather than intelligence helps reduce stigma and ensures that affected children receive appropriate, compassionate, and evidence-based support.

2.2. Literature Survey

This study examines the characteristics and detection challenges of Stuttering, a speech fluency disorder in which individuals experience involuntary disruptions during speech production, such as sound repetitions, prolongations, or speech blocks. Unlike speech delays caused by articulation or phonological difficulties, stuttering arises from complex interactions between neural, motor, and linguistic systems, resulting in irregular rhythm, disrupted fluency, and variable speech timing [1]. Previous research has explored multiple strategies for identifying stuttering, including clinician-administered fluency assessments, acoustic and prosodic analysis, and digital tools for therapy support [2].

Recent advances in artificial intelligence (AI) and machine learning (ML) have created new opportunities for automated stuttering detection and analysis. Several studies have investigated the use of acoustic and temporal features—such as speech rate, pause duration, frequency of repetitions, and pitch variability—to differentiate stuttered speech from typical fluency patterns [3]. Castillo et al. [4] demonstrated that integrating automated speech recognition systems with real-time fluency tracking can facilitate semi-autonomous speech therapy, while Sayadi [5] reported that ML classifiers trained on labeled disfluency datasets can achieve high accuracy in detecting and categorizing stuttering events.

Other researchers have focused on personalized and adaptive therapy systems, which modify fluency exercises according to individual progress and response patterns [6]. This direction aligns with the broader shift toward cloud-based and teletherapy platforms, which enhance accessibility and continuity of care, particularly for children in remote or underserved areas [7]. Clinical investigations consistently emphasize that early identification of stuttering in childhood is crucial, as timely intervention can improve fluency outcomes, reduce communication anxiety, and support long-term psychosocial well-being [8].

Overall, the reviewed literature indicates that a well-designed ML-based stuttering detection system, trained on clinically validated speech corpora and optimized for real-time processing, holds strong potential as a reliable and accessible screening tool. Such technology could complement traditional clinical assessments, enabling early risk detection, continuous progress monitoring, and data-driven therapy planning for children aged 5–10, thereby bridging the gap between conventional diagnosis and modern digital fluency interventions.

3. Research Gap

Stuttering is a speech fluency disorder that disrupts the natural flow, rhythm, and timing of speech, despite normal language comprehension and cognitive abilities. It can significantly affect a child's academic performance, self-esteem, and social participation. Although speech therapy remains the most effective intervention, early and accurate detection of stuttering is essential to maximize treatment outcomes and minimize the long-term psychosocial impact [1]. However, current methods for identifying and monitoring stuttering exhibit several key limitations.

Most existing diagnostic tools depend on manual assessment by speech-language pathologists, involving perceptual analysis of disfluencies during clinical sessions [2], [3]. While effective in controlled environments, these assessments are time-intensive, subjective, and often inaccessible to children in rural or resource-limited areas. Automated systems have been explored, but many rely on visual data (e.g., facial or articulatory movement tracking) [4], [5] or transcription-based analysis, which lacks sensitivity to subtle acoustic markers of stuttering, such as irregular pauses, tension, or pitch variation [6].

Another major limitation lies in language and context dependency. Several existing research models are developed for specific languages or datasets, limiting their generalizability across diverse linguistic and cultural populations. Furthermore, most current systems lack personalization—they do not adapt to individual differences in speech rate, emotional state, or therapy progress. This reduces their effectiveness in long-term monitoring and individualized therapy planning [7].

Finally, many available tools provide only static assessments rather than continuous progress tracking. They fail to deliver real-time feedback or predictive insights on fluency trends, which are crucial for dynamic therapy adjustment and motivation reinforcement for both children and parents [8].

To address these gaps, this research proposes the development of an audio-based, machine learning-driven stuttering detection system capable of identifying fluency disruptions from recorded speech samples without requiring visual input. By analyzing acoustic parameters such as speech rate, repetition frequency, pause duration, and prosodic rhythm, the system aims to detect early signs of stuttering in children aged 7–10. Integrated within a cloud-based monitoring platform, this solution will facilitate real-time fluency assessment, progress visualization, and adaptive therapy guidance—offering a scalable, accessible, and contact-free approach to early stuttering detection and intervention.

3.1 System Comparison

Features	[1]	[2]	[3]	[4]	Vocal Path
Audio-only detection (no visual input)	×	×	×	×	✓
Real-time classification of Stuttering	×	×	×	×	✓
Tailored for children aged 5–10	×	✓	✓	×	✓
Support Sinhala. Language	×	×	×	×	✓
Continuous progress tracking & reporting	×	✓	×	×	✓

Table 1: System Comparison

- **Stuttering Therapy Lite** – A simple Android-based app offering basic breathing and pacing exercises. It does not analyze real speech data or track progress, limiting its therapeutic usefulness [1].
- **Fluency Tracker** – Allows users to manually record stuttering frequency and triggers, but it lacks automatic detection or AI analysis, relying entirely on self-reporting accuracy [2].
- **DAF Pro** – Provides auditory feedback to assist fluent speech but requires headphones and specific speaking environments, making it impractical for children or daily use [3].
- **Stamurai (Free Version)** – Offers guided practice sessions and speech exercises but does not include AI-driven detection or adaptive personalization in its lower-tier version [4].
- The effectiveness of a mobile application "vocal path" for treated Stuttering of speech correctly [4].

4. **Research Problem**

The current landscape of stuttering detection and intervention reveals several critical gaps that limit the early identification, continuous monitoring, and effective treatment of speech fluency disorders in children. Our proposed solution aims to address these limitations by developing a comprehensive, audio-only, machine learning–based system for scalable stuttering screening and monitoring. Existing digital tools for stuttering often rely on manual clinician assessments or visual-based systems that analyze facial or articulatory movements. These methods require controlled environments, camera access, and high-quality recordings, which significantly restrict their use in home or school-based contexts [4], [5]. Moreover, many existing tools are designed for general speech therapy or rely solely on transcription-based fluency measures, neglecting the acoustic and temporal characteristics that uniquely define stuttering—such as sound repetitions, prolongations, pauses, and speech rate irregularities [3].

Another major limitation is the lack of adaptability and personalization. Most systems are language-specific and fail to adjust to variations in age, emotional state, or speaking context, making them less effective for diverse child populations [6]. Furthermore, many currently available solutions provide only static assessments without offering real-time feedback, predictive insights, or continuous tracking of therapy progress. This makes it difficult for clinicians and parents to evaluate a child’s long-term improvement or therapy effectiveness [7].

Importantly, relying solely on acoustic analysis can be misleading, as stuttering severity and frequency often vary with emotional factors, environmental conditions, and cognitive load. Therefore, an effective stuttering detection system should incorporate context-aware analysis, combining acoustic fluency features with adaptive assessment and real-time monitoring to ensure a holistic evaluation.

From these research gaps, the following questions are critical:

- How can stuttering be accurately identified using audio-only acoustic and temporal features such as repetition frequency, pause duration, and speech rate variability?
- How can the detection system be made adaptive to different languages, age groups (e.g., 5- years), and emotional contexts during speech?
- What types of machine learning models and therapy guidance mechanisms can enhance fluency outcomes and engagement for children who stutter?
- How can continuous monitoring and predictive progress tracking be integrated into a cloud-based platform to support both clinicians and families in long-term fluency management?

5. Objectives

5.1. Main Objectives

The primary objective of this research is to develop Vocal Path, a cloud-enabled, AI-driven therapeutic platform that delivers personalized and adaptive speech therapy for individuals who stutter. The system aims to integrate machine learning, speech recognition, and real-time fluency analysis within a modular framework that automates the detection of disfluencies, provides customized fluency-shaping exercises, and tracks long-term speech progress, all accessible remotely. The platform's design supports both autonomous user engagement and remote clinician supervision, thereby bridging the gap between intensive, clinic-based stuttering therapy and scalable, at-home intervention.

5.2. Specific Objectives

There are 4 Specific Objectives that must be;

1. Develop a mobile application that helps children with Stuttering of Speech with Sinhala Language Support.

This objective focuses on creating a functional mobile prototype that supports Sinhala language and provides simple, child-friendly guided speech tasks. The application will allow children to record speech in a comfortable environment and help parents and clinicians access a practical, smartphone-based tool for early fluency support

2. Implement an audio-only stuttering detection model

The objective is to build a pilot-level audio-only classifier that can identify stuttering events in children's speech. The model will focus on major disfluency categories (e.g., repetitions, prolongations, blocks) using a small dataset collected during the study. This keeps the technical scope manageable while still supporting the system's core purpose.

3. Develop a basic progress monitoring dashboard.

This objective focuses on implementing a minimal but functional progress-monitoring system. The dashboard will show key indicators, such as fluency score trends and practice frequency, enabling clinicians and parents to observe changes over time without requiring complex analytics.

4. Provide simple, rule-based pronunciation feedback and recommended exercises

This objective aims to deliver a lightweight pronunciation feedback feature and a simple personalized exercise recommender. Instead of attempting fully dynamic AI-driven adaptation, the system will match detected difficulties with pre-designed exercises, ensuring the functionality remains achievable within project constraints.

Collectively, these objectives aim to create a cloud-enabled, AI-driven speech therapy platform that promotes early detection, personalized intervention, and long-term fluency management for children affected by stuttering.

6. Methodology

6.1. Data Collection

Data collection will be conducted by the student research team, with clinical oversight provided by registered Speech-Language Therapists (SLTs) at partner institutions such as Lady Ridgeway Hospital. SLTs will act as clinical facilitators by identifying and referring eligible children from their existing caseloads, verifying the clinical suitability of each participant, and reviewing task instructions and consent materials for clinical accuracy and child safety.

Once referrals are made, the research team will take primary responsibility for approaching parents/guardians, explaining the study, obtaining written consent, conducting all recording sessions, anonymizing data, and securely uploading audio files. All procedures will strictly follow the ethical guidelines of the study and the institutional protocols agreed upon with supervising clinicians.

Each child will complete six brief tasks, each lasting approximately 20–40 seconds, allowing sufficient variation in fluency without causing fatigue. The tasks are as follows:

1. Reading Aloud (Paragraph): Reading a longer passage (30–40 words) to examine fluency during continuous speech.
2. Picture Description : Describing a short picture sequence to assess connected narrative flow.
3. Conversational Dialogue: Engaging in a short Q&A with the SLP about familiar topics such as hobbies or school activities.
4. Counting / Automatic Speech: Reciting numbers or days of the week to observe disfluencies in automatic, low-stress speech.
5. Describing a Daily Routine: Explaining a simple daily activity (e.g., “How I get ready for school”) to assess natural fluency and phrasing.
6. Free Speech / Open Topic: Speaking freely for one minute on a favorite topic to capture spontaneous fluency in unstructured contexts.

This multitask protocol provides a holistic and ecologically valid measure of speech fluency and reduces bias associated with single-task assessments such as repetition alone.

All sessions will be conducted in a quiet clinical or educational setting with the child seated comfortably. Each recording will be captured using a smartphone or tablet microphone placed approximately 15–20 cm from the speaker to ensure consistent audio quality. Short breaks (30–60 seconds) will be given between tasks to maintain focus and natural speech patterns.

Recorded audio samples will be securely uploaded to the cloud-based Vocal Path system for preprocessing and analysis. Acoustic and temporal features—including speech rate, repetition frequency, pause duration, pitch, and rhythm variability—will be extracted to identify stuttering-related disfluencies. Based on these analyses, the system will provide personalized fluency exercises such as pacing control, breathing coordination, and rhythm training activities. Real-time progress dashboards will allow both clinicians and parents to monitor changes in fluency over time.

All data collection procedures will comply with strict ethical standards, including parental consent, anonymization of child data, and secure cloud storage

6.2. Planned Data Analysis

Once we collect the voice recordings and fluency ratings, we will analyze the data carefully to identify patterns of disfluency and evaluate how effectively the Vocal Path app supports fluency improvement. First, the data will be organized into categories such as age group, stuttering severity level, and progress over time. For each child, we will calculate average fluency scores—rated out of 10 for each speech sample—based on smoothness, continuity, and speech effort. Higher averages will indicate mild or near-normal fluency, while lower scores will reflect more frequent or severe stuttering behaviors, such as repetitions, prolongations, and blocks.

We will then compare groups—children who stutter versus those with typical fluency—to identify differences in features like pause duration, repetition frequency, and speech rate variability. Descriptive statistics such as means, ranges, and standard deviations will help determine whether the app accurately detects and classifies stuttering severity. To measure therapeutic progress, we will track each child’s fluency scores from the start of therapy to later sessions, observing whether scores improve—for instance, from 4 to 7 after several weeks of targeted practice. Line charts and progress graphs will be used to visualize these score trends clearly.

To assess the app’s predictive accuracy, we will use past fluency data to forecast expected future performance and then compare those predictions to the actual post-therapy results. Metrics such as mean absolute error (MAE) or root mean square error (RMSE) will indicate how close the predictions are to real improvements—lower errors meaning better reliability. Additionally, feedback from speech therapists will be collected to validate whether the app’s analytics align with their clinical observations. If users consistently demonstrate higher fluency scores and therapists confirm noticeable progress, it will support the app’s effectiveness as a stuttering therapy aid.

All analyses will be conducted under strict data privacy and ethical protocols, ensuring that no personal information is shared. This approach provides a robust,

evidence-based framework to evaluate the system's accuracy, clinical usefulness, and potential for improving speech fluency and confidence in children who stutter

6.3. Reliability and Validity Measures

6.3. 1 Participant Recruitment and Approach

Participants will be recruited via referrals from registered Speech-Language Therapists (SLTs) and certified child-speech therapy clinics, ensuring potential participants are identified within professionally supervised clinical settings. SLTs will act as clinical facilitators by confirming eligibility and providing an initial referral

Following referral, a member of the student research team will make first contact with parents or guardians to provide study information, answer questions, and arrange an in-person meeting in a structured, ethically controlled setting. The research team will be responsible for explaining the study, obtaining written informed consent, and managing all administrative aspects of enrollment. The supervising SLT will be available to provide clinical clarification or to join the initial meeting if parents request additional reassurance.

To protect child welfare and maintain clinical standards, the following safeguards will be observed:

- Only children aged 5–10 years will be invited to participate.
- Written informed consent will be obtained from parents or legal guardians; verbal assent will be sought from each child using age-appropriate language.
- No child will be approached or recorded by research staff without prior referral and without parental consent.
- The SLT may be present or on-call during recruitment or data-collection sessions upon request

This recruitment pathway balances ethical oversight from clinical partners with clear research ownership by the investigators, ensuring participant safety, comfort, and procedural transparency.

6.3. 2 Study Set & Setting

Data collection for this study will be conducted in a quiet, private, and child-friendly room located within a registered clinical or therapeutic environment that is routinely used for speech and language assessments. The setting will be selected to ensure that children feel safe, relaxed, and comfortable throughout the session. To maintain high-quality audio recordings, the room will be arranged to minimize external noise, avoid echo, and reduce interruptions. A comfortable seating arrangement will be provided for both the child and the accompanying parent or guardian, who will remain present or within visible proximity at all times to offer reassurance and emotional support.

The environment will be arranged in a way that feels familiar to children, using simple visuals or child-appropriate materials to reduce anxiety and encourage natural speech. Only essential individuals will be present during the session to avoid distraction. All necessary recording equipment will be set up in advance to prevent delays and ensure a smooth workflow. For ethical and privacy reasons, no data collection will occur in public areas, school classrooms, waiting rooms, or open clinical wards, ensuring that confidentiality and child safety are maintained consistently throughout the research process.

6.3. 3 Communication With Children and Parents

Communication with children and parents will be conducted with sensitivity, clarity, and in strict accordance with ethical guidelines. The research team will deliver all study instructions and explanations in plain, age-appropriate language. Prior to data collection, the SLT will be available to provide additional clinical clarification or reassurance to families on request.

Children will be addressed using simple, friendly, and developmentally appropriate language so tasks are understandable without pressure. Parents or guardians will receive:

- a written information sheet explaining the study purpose and procedures;
- a clear verbal overview of the child's involvement delivered by a member of the research team; and
- a written consent form outlining:
 - the purpose of the study;
 - the specific tasks the child will complete;
 - the approximate time required;
 - measures to protect privacy, data security, and anonymity; and
 - their right to withdraw at any time without consequence.

Verbal assent will be sought from each child by a trained member of the research team using child-friendly language. Participation will be explicitly presented as voluntary; if a child expresses discomfort or reluctance, the session will be paused or discontinued.

6.3. 4 Number of Participants and Speech Samples

The study plans to recruit approximately 50–80 child participants children who stutter. This sample size is considered feasible within a clinical setting and aligns with recommended guidelines for pediatric speech research, ensuring adequate variability in speech patterns, age, and fluency characteristics. Each child will be asked to complete 6 short and structured speech tasks, and taken around 10-12 speech samples from a child patient which may include activities such as reading aloud, picture description, conversation prompts, and simple automatic speech tasks. These tasks are intentionally brief and child-friendly to reduce fatigue and maintain engagement.

Across all participants, the study is expected to generate 500–800 speech samples, providing a sufficiently large dataset for both statistical analysis and machine learning model development. This dataset size is chosen to:

- Support statistically meaningful comparisons across age groups and fluency levels
- Ensure diversity in linguistic content and speech contexts, which improves model generalizability
- Provide enough labelled examples to train, validate, and test the ML classifier without overfitting
- Maintain ethical feasibility by not requiring excessive task repetition from any single child

Overall, this structured and balanced sampling approach ensures that the collected data will be both clinically representative and methodologically sound.

6.3. 5 Sample Size Justification and Power Analysis

The sample size for this study has been determined based on established research standards for child-speech classification tasks and the statistical requirements needed to ensure valid and reliable outcomes. In machine-learning studies involving pediatric speech, a minimum of 300–500 labeled audio samples is typically required to achieve stable training performance and reduce the risk of model overfitting. Considering the study design in which each child will contribute ten short speech tasks, recruiting 50–80 participants provides a dataset of approximately 500–800 recordings, meeting and exceeding this minimum requirement.

From a statistical perspective, meaningful comparisons of fluency behaviors across multiple task types generally correspond to a medium effect size (Cohen’s $d \approx 0.5$), which is widely observed in prior child fluency research. To detect such an effect reliably, power analysis using standard tools such as G*Power recommends the following parameters:

- Statistical power of 0.80 ($1-\beta$)
- Significance level $\alpha = 0.05$
- Two-group comparisons or repeated-measures frameworks

Under these assumptions, power analysis indicates that approximately 30–35 participants per group are needed, resulting in a total recommended sample of 60–80 children. Therefore, the proposed recruitment range of 50–80 child participants is both statistically

justified and practically feasible, ensuring sufficient power for group-level analyses while simultaneously providing enough labeled speech data for robust machine-learning training and validation.

6.3. 6 Duration and Structure of Participation

Each child will participate in a single data-collection session designed to be brief, comfortable, and developmentally appropriate. Sessions are expected to last approximately **15–20 minutes** in total and will include optional short breaks to minimise fatigue and maintain attention. The research team will conduct the session and manage pacing; the supervising Speech-Language Therapist (SLT) will be available for clinical consultation or to join the session on request.

Individual speech tasks are planned to last **about 30–60 seconds** each, depending on the child’s speaking rate and familiarity with the activity. Each child will complete six structured or semi-structured tasks to capture a broad range of natural fluency behaviours. Task types include:

- reading short, age-appropriate words or sentences (where literacy permits)
- describing simple pictures or illustrated scenes
- responding to short conversational prompts or guided questions
- producing automatic speech sequences (for example: counting, days of the week)
- retelling a brief, familiar story or describing a routine

Short pauses or breaks will be offered between tasks, and children may skip, repeat, or modify any task according to their comfort. A parent or guardian will remain nearby during the session to provide reassurance. This structured yet flexible approach aims to collect high-quality speech samples while prioritizing each child’s well-being and comfort.

6.3. 7 Benefits to Participants

Participation in this study does not provide any direct clinical or therapeutic benefit to the children or their parents. The speech activities conducted are designed purely for data collection and system development purposes and are not intended as a substitute for formal speech-language therapy or medical intervention. Children will not receive immediate fluency improvement, diagnosis, or individualized therapy plans as a result of participating.

However, participants’ involvement is highly valuable for the advancement of local speech technology and public health resources. The data gathered will contribute to the development of an AI-based, Sinhala-language speech analysis system, designed to identify and monitor stuttering behaviors more accurately in future clinical settings. Over time, this system aims to support Speech-Language Therapists (SLTs) across Sri Lanka by providing objective fluency measurements, helping reduce the subjectivity and time burden of manual assessments.

Parents and children who participate are thus making a meaningful contribution to the improvement of early screening tools and future clinical care for children who stutter. The

broader Sri Lankan community will ultimately benefit from increased awareness, improved access to diagnostic tools in underserved areas, and a growing national database on childhood fluency patterns.

All participants will be fully informed that their role in this study is voluntary, non-therapeutic, and primarily contributes to research aimed at improving future speech therapy technology rather than providing direct personal benefit.

6.3. 8 Potential Risks or Discomforts

The risks associated with participation in this study are minimal and primarily related to the speaking tasks involved. Some children may experience **mild fatigue** when completing multiple tasks in a short period, especially if they are not accustomed to structured speech activities. A small number of children may feel **minor embarrassment, hesitation, or frustration** if they find certain speech tasks difficult or if they are sensitive about their speech fluency. Additionally, younger participants may experience **boredom or attention fatigue** if their focus naturally declines during the session.

It is important to emphasize that there are **no physical, medical, or invasive procedures** involved in this research. The tasks are similar to those commonly used during routine speech-therapy sessions and pose no risk to the child's physical safety or emotional well-being. The presence of a trained Speech-Language Therapist ensures that any signs of discomfort are immediately addressed, and the session can be paused or terminated at any point. Overall, the likelihood and severity of risk are considered very low, falling well within acceptable ethical standards for child speech research.

6.3. 9 Measures to Minimize Inconvenience

The study includes several proactive measures to ensure children experience minimal inconvenience and remain comfortable throughout participation. Activities are presented as short, playful speech tasks rather than formal examinations, helping children feel relaxed and reducing performance pressure. The research team will monitor the child's comfort and engagement during each session and will pause, offer reassurance, or discontinue the session if the child appears tired, anxious, or unwilling to continue. A supervising Speech-Language Therapist (SLT) will be available for clinical advice or to join the session on request.

Children will be free to take breaks at any time to rest, drink water, or reset emotionally. The pace and sequence of tasks will be flexible and tailored to each child's needs. A parent or guardian will remain nearby throughout the session to provide familiarity and emotional support.

Audio-only recording is used to reduce stress and preserve privacy; no video or photographs will be taken. This approach reduces feelings of being observed and lowers the risk of participant identification.

Additional measures to minimise inconvenience:

- keep sessions brief (under 20 minutes where possible);
- permit children to skip, repeat, or adapt tasks according to comfort;
- use child-friendly visual aids and simple instructions to maintain engagement;

- prepare the testing space in advance to reduce waiting time; and
- limit the number of people present during sessions to avoid distraction.

All procedures prioritize the child’s autonomy, emotional well-being, and safety and will be carried out in accordance with approved ethical guidelines.

6.3. 10 Data Collection Protocol

Data collection will follow a tightly controlled, secure, and ethics-compliant protocol designed to protect the privacy and well-being of all child participants. Only **audio recordings** will be captured, using high-quality, clinically approved recording equipment appropriate for speech assessment. No video, photographs, or other identifiable images will be recorded at any stage to minimise participant discomfort and protect privacy.

Immediately after each session the research team will assign a unique anonymized identifier to the audio file; this identifier will not contain the child’s name, school, or other personal information. The mapping between identifiers and participant details will be held separately in an encrypted file accessible only to authorised personnel — specifically the supervising Speech-Language Therapist (SLT) and named members of the research team — and will be used solely for approved administrative or clinical safety purposes.

Anonymized audio files will be uploaded to a secure, encrypted, access-controlled storage system that meets institutional data-protection requirements for research involving children. Access controls will prevent unauthorized retrieval or sharing of sensitive material; only designated project staff and supervising clinicians will be permitted to access the recordings for analysis and annotation.

Additional safeguards include:

- pre-testing and routine checks of recording equipment to avoid repeat sessions;
- secure, versioned backups to prevent accidental data loss;
- prohibition of transferring raw data to personal devices; and
- detailed session logs maintained for audit and quality-assurance purposes.

These measures ensure that audio data are collected, stored, and processed with the highest standards of privacy, security, and ethical responsibility

6.3. 11 Education and Guidance for Parents

The research team will ensure that parents and guardians receive clear, comprehensive information about the study so they can make an informed decision before providing consent. At the start of the recruitment process the research team will provide an accessible verbal overview of the study’s aims and procedures in plain language. If parents or guardians would like a clinical perspective, the supervising Speech-Language Therapist (SLT) will be available to offer a brief explanation of childhood stuttering as a developmental fluency disorder, common speech behaviours, and the importance of early assessment.

Families will be informed how participation contributes to improving speech-assessment tools — specifically the development of digital methods intended to support earlier detection, ongoing monitoring, and more accessible intervention for children with fluency

difficulties. The information provided will also clearly explain how the child's audio data are handled: files will be anonymized, stored securely, and accessible only to authorised project personnel.

Parents and guardians will receive a written information sheet summarising:

- the purpose of the study and the procedures to be followed
- the types of speech tasks the child will be asked to perform
- the total expected time commitment for participation
- the measures in place to protect privacy and confidentiality (including anonymization and storage safeguards)
- their right to withdraw at any time without affecting the child's clinical care or services

Before signing the consent form parents and guardians will be encouraged to ask questions and seek clarification. The research team will ensure that all questions are answered fully; the supervising SLT will be available for additional clinical clarification on request. Written informed consent will be obtained by a member of the research team and documented according to institutional ethical requirements

6.4 Ethical Conduct of Study-

Ethical compliance forms the foundation of this study, ensuring that all procedures are conducted responsibly and in accordance with globally recognized principles of research involving human participants. The research involves children aged 5–10 years, a vulnerable population that requires heightened ethical sensitivity. This section outlines how the principles of informed consent, beneficence, non-maleficence, justice, autonomy, and privacy are systematically integrated into the study’s design, data collection, and dissemination processes. These measures guarantee that participants’ rights, safety, and dignity are protected at every stage of the research process. The study adheres to institutional ethics guidelines and aligns with standards set forth by the Sri Lanka Institute of Information Technology (SLIIT) Research Ethics Committee and international pediatric research frameworks such as the Declaration of Helsinki and Belmont Report.

Informed Consent-

Obtaining valid and meaningful informed consent is a critical ethical requirement in this study. Prior to recruitment, parents or legal guardians will receive comprehensive information sheets explaining the study’s purpose, objectives, procedures, potential risks, data handling, and expected time commitment. This explanation will be provided both verbally and in writing, using clear and culturally appropriate language in Sinhala, Tamil and English to ensure full understanding. Parents or guardians will be encouraged to ask questions and will only sign the consent form after confirming they understand all aspects of the study. For child patients, verbal assent will be obtained through simplified explanations that match their developmental level, ensuring they comprehend what will happen during the speech tasks. This dual consent–assent approach ensures voluntary participation by both guardian and child. Copies of signed consent forms will be securely stored, and participants will be reminded that participation is entirely optional. No incentives or coercive elements will be used to influence their decision, and withdrawal will be permitted at any time without any adverse consequence.

Beneficence and Non-Maleficence -

The principle of beneficence emphasizes maximizing benefits and minimizing potential harm to participants. This study is low-risk and poses no known physical or psychological hazards. The speech-based tasks are short, non-invasive, and designed to mimic natural conversational or reading activities typically conducted in speech-language therapy. Sessions will be carried out in familiar, comfortable environments such as clinics, reducing stress and ensuring the child feels at ease. The non-maleficence aspect ensures that no harm will come to participants through undue emotional pressure, fatigue, or data misuse. Although there is no direct therapeutic or financial benefit for participants, their contributions have a substantial societal benefit — they will help advance Sinhala-language stuttering analysis and contribute to developing a culturally adapted digital platform that supports early diagnosis and monitoring for future children. Any signs of distress or discomfort will immediately lead to a pause or termination of the activity, with appropriate reassurance provided by the supervising speech-language therapist. This proactive attention to well-being ensures that participation remains safe, respectful, and child-centered.

Justice -

The principle of justice ensures fairness in participant recruitment, inclusion, and data handling. All participant selection will be conducted equitably, without discrimination based on gender, ethnicity, religion, or socioeconomic background. Recruitment will be facilitated through registered Speech-Language Therapists (SLTs) who identify suitable participants based on clinical need and age criteria. Participation will be voluntary and open to both male and female children who meet the study inclusion standards. The study will avoid overburdening any single group and ensure that benefits and knowledge gained are shared broadly with the wider community. No participants will be excluded unjustly, and every effort will be made to include children from diverse linguistic and social backgrounds to ensure fair representation. The research team is committed to adhering to the ethical principle of distributive justice, meaning that any advantages resulting from the project (e.g., development of improved tools for assessment) will ultimately serve the public good rather than specific individuals. Moreover, all research outcomes will be reported transparently, without selective omission or manipulation of data, ensuring academic honesty and fairness.

Autonomy-

Respecting autonomy means acknowledging and protecting each participant's right to make informed, voluntary decisions about their involvement in the research. In this study, autonomy applies both to parents and to child participants. Parents or legal guardians have the authority to provide or withhold consent, while children are encouraged to express their willingness through verbal assent. At any point during the study, participants may choose to withdraw without facing negative consequences or affecting their access to current or future therapy. To safeguard autonomy, participants will be informed that their decision to participate—or not—will remain confidential and will not influence their relationship with their therapist, school, or research team. During each session, the child's emotional state and willingness to continue will be continuously monitored. If a participant exhibits discomfort, fatigue, or anxiety, the session will be paused or stopped entirely. This adaptive, child-focused approach respects individual choice, psychological comfort, and personal dignity. Ultimately, respecting autonomy ensures that participation remains truly voluntary, empowering both parents and children as active, informed partners in the research process.

Privacy and Confidentiality-

Maintaining participant privacy and ensuring data confidentiality are paramount ethical obligations of this research. All data collected, especially voice recordings—will be handled according to stringent data protection standards. Personal identifiers such as names, addresses, or contact details will never appear in data files used for analysis. Each participant will instead be assigned a unique coded ID immediately upon enrollment. Recorded audio files will be encrypted and stored on secure, password-protected servers with restricted access. Only authorized members of the research team and supervising clinicians will have access to the original, identifiable data. Anonymized datasets may be used for statistical analysis or academic publications, but all identifying information will be permanently removed. No audio samples will be shared publicly or used for commercial purposes. In accordance with Sri Lankan Data Protection Act No. 9 of 2022, all digital records will be retained for the minimum necessary duration and destroyed securely afterward. This comprehensive approach ensures that the privacy, dignity, and trust of participants and their families are safeguarded throughout and beyond the study period.

System Diagram

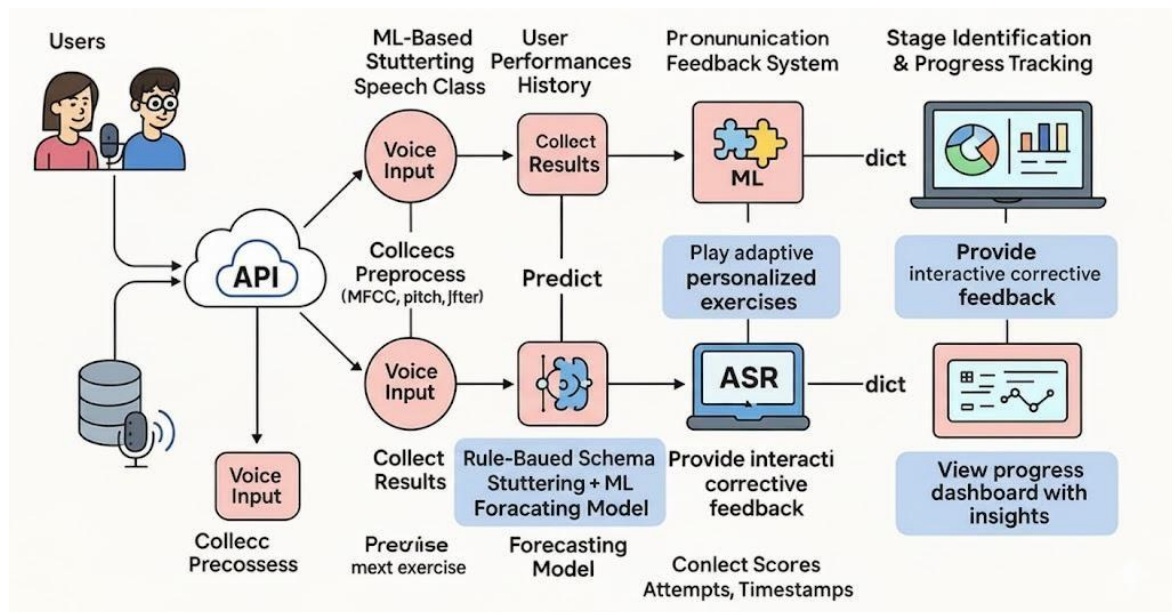


Figure 1 : System Diagram

Figure 01 illustrates the overall high-level architectural diagram of the proposed system. In the proposed system, users provide **voice input** which is collected and preprocessed using acoustic features like MFCC, pitch, and jitter via the cloud-based **API**. This preprocessed input is then fed into an **ML-based Stuttering Speech Class** (classifier), incorporating user history and performance data from the database to collect results and predict **Stuttering** presence. After detection, the system directs users to pronunciation feedback through adaptive personalized exercises via a Pronunciation Feedback System and provides interactive corrective feedback. Simultaneously, the voice input is processed via **ASR** (Automatic Speech Recognition) to collect results, which are analyzed using a **Rule-Based Schema Stuttering + ML Forecasting Model** to predict the next exercise and provide feedback on concepts, scores, attempts, and timestamps. These adaptive exercises and progress tracking help to reduce the impact of **Stuttering** by monitoring stages, providing insights via dashboards, and enabling continuous improvement.

Components System Diagram

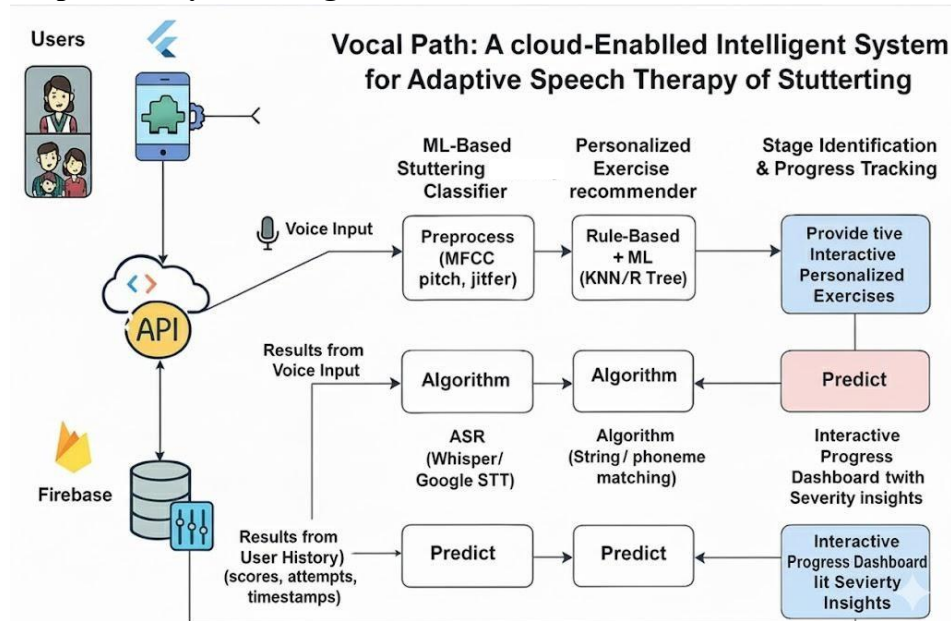


Figure 2: Component System Diagram

This figure 02 illustrates the architecture of **Vocal Path**, a cloud-enabled intelligent system designed for the adaptive speech therapy of Stuttering . The system is comprised of three primary processing pipelines that work together to analyze user input, recommend exercises, and track progress.

Users interact with the system via a mobile application, which captures their **voice input** and sends it to a backend API. The first pipeline processes this voice input through an **ML-Based Stuttering Speech Classifier**. Here, the audio is preprocessed to extract key vocal features like MFCC, pitch, and jitter. These features are then fed into a **Personalized Exercise Recommender** that uses a combination of rule-based logic and machine learning models (like KNN/R Tree) to identify the user's current stage. This allows the system to provide interactive and personalized exercises tailored to the user's specific needs.

The second pipeline uses an **Automatic Speech Recognition (ASR)** engine, such as Whisper or Google STT, to convert the user's speech into text. A subsequent algorithm then performs string and phoneme matching to evaluate the accuracy of the user's pronunciation. The results from this analysis are used to predict the severity of the speech disorder and populate an **Interactive Progress Dashboard** with detailed insights.

Finally, the third pipeline leverages historical data, such as scores, attempts, and timestamps, which are stored in a **Firestore** database. This data is fed through predictive models to analyze long-term trends and user improvement over time. The output of this pipeline also contributethe **Interactive Progress Dashboard**, providing a comprehensive overview of the user's journey.

7. Project Requirements

7.1. User Requirements

❖ Patients

- An **intuitive and accessible interface** that allows patients to easily navigate and engage with speech exercises and therapy activities.
- **Clear, simple instructions** for each exercise, provided through both visual and auditory cues to ensure patients can use the application independently.
- Access to a **personal progress dashboard** that provides immediate, easy-to-understand feedback on their performance and tracks improvement over time to keep them motivated.

❖ Consultants (Therapists/Clinicians) –

- A comprehensive dashboard to **manage multiple patients**, view their profiles, and monitor their overall progress efficiently.
- Access to **detailed analytical reports** summarizing patient performance, including severity scores, accuracy metrics, and long-term progress trends.
- The ability to **remotely customize therapy plans** by assigning specific exercises and adjusting difficulty levels based on the data-driven insights from the system.

7.2. System Requirements

❖ Hardware Requirements.

➤ Mobile Devices (For Users):

- iOS Devices: iPhone or iPad running iOS 13 and above.
- Android Devices: Smartphones or tablets with Android 5.0 (Lollipop) and above.
- Enough memory and storage space for smooth application performance

➤ Computers (For Development and Testing):

- Operating System: Windows 7 and above, macOS, or Linux.
- Processor: Dual-core 2 GHz or equivalent.
- RAM: 4 GB or higher.
- Storage: SSD recommended for faster development.

❖ Software

➤ For Design and Development:

- Flutter (latest stable version) – for building cross-platform mobile apps
- Dart SDK (latest stable version) – required to run Flutter apps
- Integrated Development Environment (IDE): Visual Studio Code or Android Studio
- Android SDK / Android Studio – for building and testing on Android
- Xcode (if developing for iOS, on macOS) – for building and testing on

- iOS
 - Flutter CLI – for running, debugging, and building Flutter projects

7.3. Functional Requirements

- ❖ **Collect a comprehensive dataset** of children with both Stuttering -affected and non-affected speech samples to ensure the dataset remains balanced. This helps the system achieve higher accuracy and reliability in classification.
- ❖ **Develop a machine learning–based classifier** to predict whether a child is at risk of Stuttering of Speech, ensuring the model provides consistent and clinically valid results.
- ❖ **Validate the deployed API** by testing it with multiple real-world speech samples, making sure it functions as expected and delivers reliable, real-time classification outputs.
- ❖ **Provide child-friendly interactive exercises** and feedback sessions that encourage children to practice speech production in a motivating and engaging way, supporting therapy alongside detection.
- ❖ **Implement a progress monitoring system** to track improvements over time and generate structured reports, so that parents, teachers, and clinicians can clearly see the child’s development and therapy outcomes.

7.4. Non-Functional Requirements

- ❖ **Usability:** To help users with Stuttering , the program should be simple to use, with clear instructions and intuitive navigation.
- ❖ **Accessibility:** For users with disabilities to benefit from the software, accessibility requirements must be followed. This includes having choices for high contrast colors, larger font, and screen reader compatibility.
- ❖ **Performance:** To provide a seamless user experience, the app should be snappy and powerful even on devices with less processing power.
- ❖ **Security:** The app should adhere to security best practices to protect user data and ensure privacy.
- ❖ **Maintainability:** The app should be easy to maintain and update, with modular code

and clear documentation.

- ❖ **Performance Metrics:** The app should include metrics for tracking user progress and engagement, allowing for continuous improvement and optimization of the app's effectiveness in addressing Stuttering

8. Budget and Budget Justification

The below table 08 depicts the overall budget of the entire proposed system

Component	Amount (USD)	Amount (LKR)
Traveling fee ▪ Doctor consultation. ▪ Data set gathering.	19.20	6000.00
Internet charges ▪ Development. ▪ Technical Information Learning.	9.60	3000.00
Online resources and Subscriptions ▪ APIs. ▪ Cloud services. ▪ Databases.	32.00	10000.00
Total	60.80	19000.00

Table 2: 1.Budget and Budget Justification

- **Doctor Consultation & Data Gathering** – Required visits to Hospitals to meet Speech-Language Pathologists (SLPs) and collect children’s speech samples. Minimal transport and consultation fees are included.
- **Internet Charges** – Used for development, communication, and accessing online tutorials.
- **Online Resources & Subscriptions** – Covers essential cloud services such as Firebase hosting, database storage, and limited API usage for speech processing. Most free-tier options will be utilized to minimize costs.

❖ Research Timeline Grant Chart -

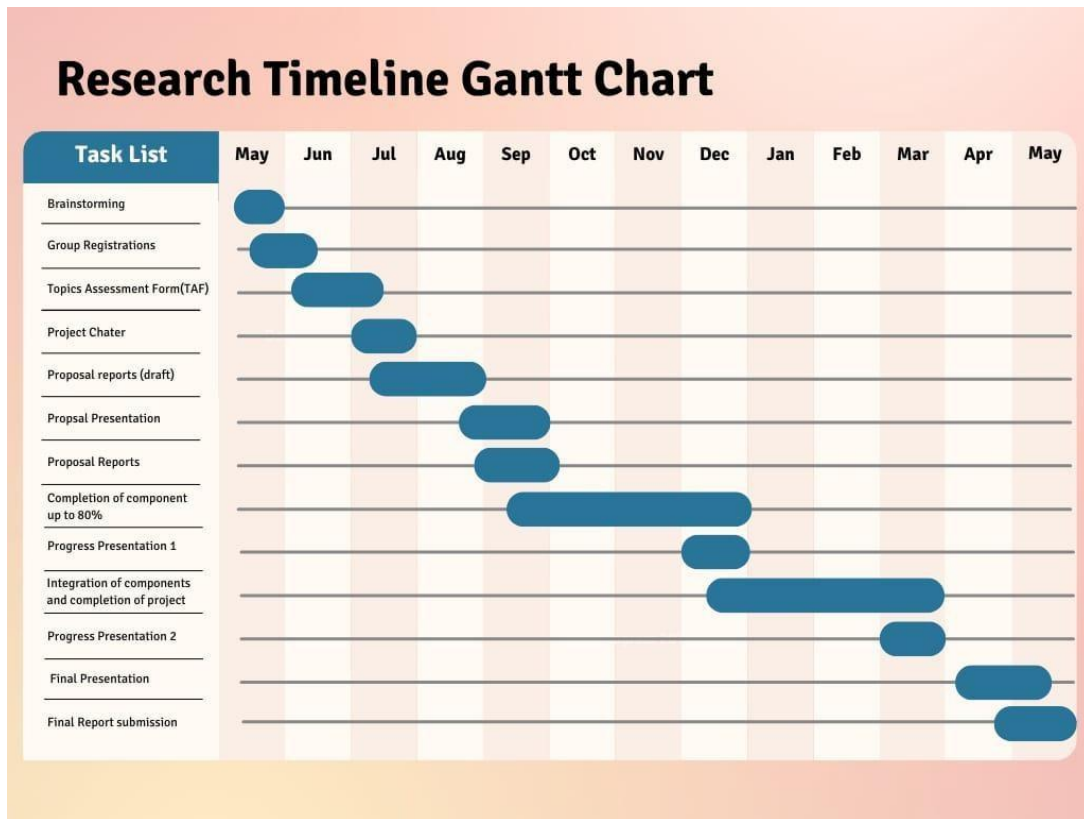


Figure 3 research Grant Chart

❖ Workflow Breakdown Structure -

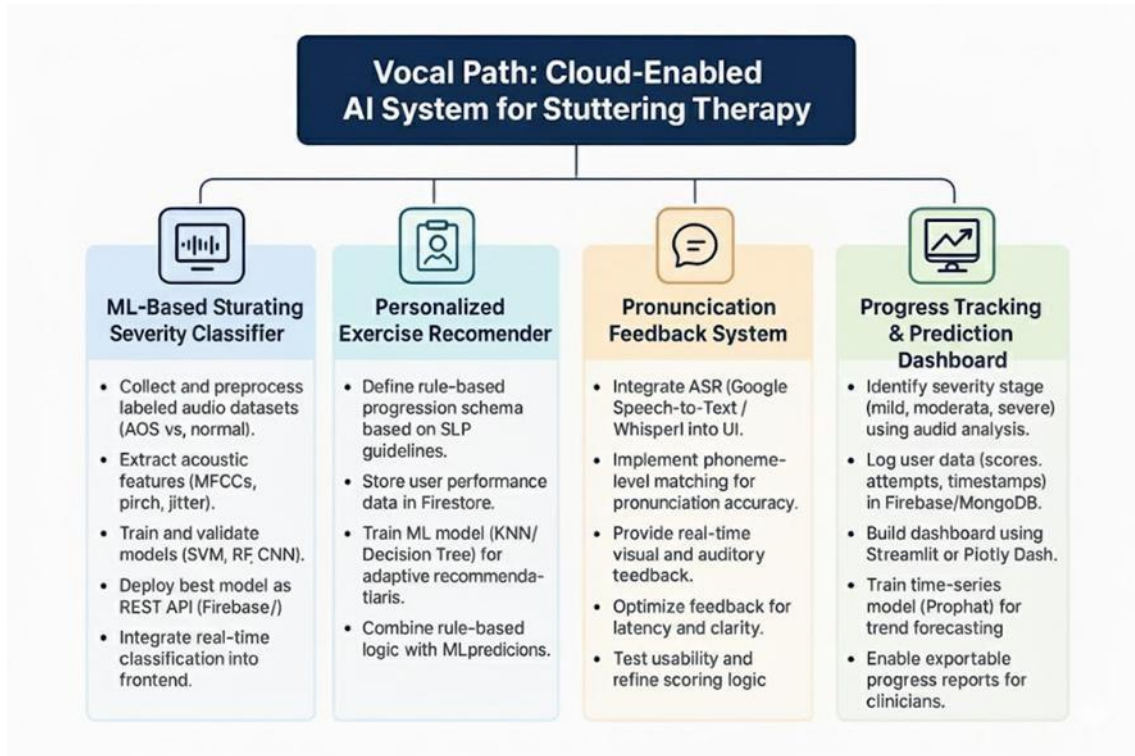


Figure 4 workflow Breakdown Structure

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